

Problem

A series-tuned antenna circuit consists of a variable capacitor (40 pF to 360 pF) and a 240- μ H antenna coil that has a dc resistance of 12 Ω .

(a) Find the frequency range of radio signals to which the radio is tunable.

(b) Determine the value of Q at each end of the frequency range.

Step-by-step solution

Step 1 of 2

$$(a) f_o = \frac{1}{2\pi\sqrt{LC}}$$

when $C = 300PF$

$$f_o = \frac{1}{2\pi(240 \times 10^{-6})(300 \times 10^{-12})} = 0.541 MHz_z$$

when $C = 40PF$

$$f_o = \frac{1}{2\pi(240 \times 10^{-6})(40 \times 10^{-12})} = 1.624 MHz_z$$

There fore the frequency rance is

$$0.541 MHz_z < f_o < 1.624 MHz_z$$

Step 2 of 2

$$(b) Q = \frac{2\pi fL}{R}$$

at $f_o = 0.541 MHz_z$

$$Q = \frac{(2 \times 3.14)(0.541 \times 10^6)(240 \times 10^{-6})}{12} = 67.98$$

at $f_o = 1.624 MHz_z$

$$Q = 204.1$$